

Question	Answer	Marks
2(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
2(b)	<i>Any two points from:</i> • molecules are in <u>continuous</u> random motion • molecules have negligible volume compared with volume of gas • collisions (involving molecules) are (perfectly) elastic • collisions (of molecules) are instantaneous	B2
2(c)(i)	$pV = nRT$	C1
	$p = (0.0160 \times 8.31 \times 282) / (1.87 \times 10^{-4})$ $= 2.01 \times 10^5 \text{ Pa}$	A1
2(c)(ii)	number of molecules = $0.0160 \times 6.02 \times 10^{23}$	C1
	separation = $\sqrt[3]{[(1.87 \times 10^{-4}) / (0.0160 \times 6.02 \times 10^{23})]}$ $= 2.7 \times 10^{-9} \text{ m}$ (<i>allow any answer that is $3 \times 10^{-9} \text{ m}$ to one significant figure</i>)	A1
2(d)(i)	$F = 6.67 \times 10^{-11} \times (3.34 \times 10^{-27})^2 / (2.7 \times 10^{-9})^2$	C1
	$= 1.0 \times 10^{-46} \text{ N}$	A1
2(d)(ii)	numerical comparison between 10^{-46} N (F) and 10^{-26} N (the weight of molecule) leading to a conclusion that the assumption is supported	B1

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